Journal of Molecular Liquids 370 (2023) 121034

Contents lists available at ScienceDirect

Journal of Molecular Liquids

journal homepage: www.elsevier.com/locate/molliq

Suspended water nanodroplets evaporation and its deviation from continuum estimations

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ARTICLE INFO

Article history:
Received 5 July 2022
Revised 5 December 2022
Accepted 7 December 2022
Available online 11 December 2022

Keywords:
Molecular dynamics
Water nanodroplets
Nanodroplets evaporation
Evaporation
Data-driven model
Evaporation rate

ABSTRACT

Although droplets evaporation is fundamental to many applications and phenomena, accurate prediction of their evaporation trajectory is still a technological gap today. This is even more so for nanodroplets when compared to the macro-sized droplets. Since it is technically challenging to observe real nanodroplets evaporation today, this contribution uses full atomistic molecular dynamics simulations to observe and quantify suspended water nanodroplets evaporation. There are two main findings to this contribution. First, it is shown in detail that continuum analysis significantly over predicts the evaporation rate of nanodroplets. This makes continuum analysis unsuitable to predict water nanodroplets' behavior. Secondly, this contribution illustrates the use of data-driven approach that can be used for nanodroplets evaporation behavior prediction, instead of running full atomistic molecular dynamics simulations for every different condition, which can be prohibitively expensive. The paper hopes to encourage scientists working in similar field to share their MD results for nanodroplets evaporation to build a data-driven model that could be used to predict nanodroplets evaporation trajectory at a fraction of the cost of full-scale MD simulations.

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1. Introduction

Droplets evaporation is fundamental to many applications and phenomena, examples include spray drying, combustion and design of nanodevices [1–4]. The understanding of droplets behavior is also crucial to health and safety assessments involving toxic chemical releases [5–6]. Evaporation of droplets greatly impacts the behavior of droplets as they disperse through the environment [7–8], and hence, have a direct impact on the dynamic change in size of droplet and their eventual trajectory. A good understanding of their evaporation behavior is therefore the key to control and manipulation, as well as predicting phenomena outcome involving droplets. The ability to accurately model the evaporation of droplets came under the limelight with the ongoing Covid-19 pandemic, where the accurate prediction of evaporation behavior of the droplets impacts the accurate prediction of trajectory and lifetime of airborne viral containing droplets [9–11]. Droplets can exist across multiple length scales and can transcend between the macro to nano length scales as the droplets evaporate or break up, or vice versa through coalition with other droplets.

Today, evaporation of nano sized droplets is much less explored when compared to the macro-sized droplets. This is due mainly to the difficulty in observing the process of evaporation using experimental means, especially so in the nanoscale [12]. For macro-sized droplets of size microns and above, state-of-the-art cameras can be utilized to observe the evolution of evaporation. Niimura and Hasegawa [13] for example used the single-axis acoustic levitator and a high-speed camera to observe the evaporation process of droplets made up of different chemical component. Through their observations, they developed equations for evaporation of binary droplets. Chaudhuri et al. in 2021 used a charge coupled device (CCD) camera to observe evaporation of 300 μm droplets and used their findings to compare with simulated and theoretical results [14]. While droplets of more than a few hundred microns can be observed, evaporation for droplets in the nanometer scale is still challenging to observe with current technology. State-of-the-art cameras are also unable to resolve to the molecular length scale, where the evaporation process essentially takes place [15]. In lieu of these limitations, Molecular dynamics (MD) simulations fill this gap. MD is a simulation technique that models individual atoms and their interactions with other atoms. It computes the forces on these atoms at every time step based on the atomic model input, and subsequently resolves the kinematics of these atoms

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https://doi.org/10.1016/j.molliq.2022.121034
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at every time step [16]. As MD do not require many simplifying assumptions to be made, it is widely used to study and observe molecular phenomenon, which are currently still very challenging and costly in physical experiment [17].

Some of the early works that used MD to study the behavior of nanodroplets include the work by Long et al in 1996 [18], and Landry et al in 2007 [19]. Due to the lack of advanced potential models that describe liquid behaviors, such as TIP4P [20] or ReaxFF [21] amongst others, early works tend to use simple Lennard-Jones potential with simple droplets structure, such as all argon nanodroplets. However, since then, there are more literature in recent years using more advanced potentials and modelling techniques, which suggest that nanodroplets behavior is different from continuum estimation [15,22–23]. In particular, Wang et al. in 2013 reported that there is a general deviation from classical d2-law to d-law for nano-droplets [24]. This same phenomenon was observed by Pestana and Head-Gorden [25] for evaporation of water nano-droplets on heated surfaces. In 2017, Hołyst et al. reported that there exists a temperature discontinuity a distance A λ from the droplet-vapor interface, reducing heat flux of droplets by a factor of $\frac{R}{R+A\lambda}$ due to the switch between continuum heat diffusion to ballistic heat transport [15]. In 2019, Rana et al. illustrated using a temperature-jump boundary condition to modify current continuum laws to predict the temporal evolution of nano-droplet diameter through evaporation process (3). These works ascertain that the evaporation process at the nanoscale may not behave like the continuum scale. However, there is a lack of MD simulations studying the volumetric change due to evaporation of water nanodroplets at different conditions. Such information can be valuable for engineering applications or scientific predictions [13,26]. The current work aims to address this gap and discuss the validity of using continuum classical diffusion theory to predict the evaporation behavior of nanosized water droplets at different ambient temperatures. In this work, MD simulations are performed for nano-sized water droplets evaporation to document the behavior of water nanodroplets evaporation for different initial size and temperature conditions. It is also clearly shown in this work that the evaporation rate for water nanodroplets deviates from continuum predictions significantly.

This work also outlines a data-driven methodology to help researchers and scientists estimate evaporation rates of water nanodroplets of varying sizes and temperature conditions without running computationally expensive MD simulations. This is valuable, as it can be computationally expensive to run a MD simulation to predict the evaporation rate every single time the temperature or the size of the droplets change. The data will also be made publicly available for the academic community to supplement, or validate, to work towards using data-drive methods to supplement the expensive MD simulation runs (url: https://github.com/elisaaym/Water_nanodroplets_evaporation). Data-driven methods to supplement or replace costly and time-consuming simulations are becoming a reality with advancement in machine learning and data science [27–28]. In this contribution, it is further shown that the prediction of evaporation rates of nanosized droplets under different conditions can be performed reasonably accurately with data-driven methods, at a fraction of the cost of MD simulations.

In the following sections, the MD and continuum calculation methodology will first be presented. The MD results and their deviations from continuum calculations are presented and discussed next. Lastly, a data-driven approach to predict the evaporation rate of water-nanodroplets at different temperatures is presented and its accuracy discussed. As far as the authors are aware of, this is the first work to systematically explore the evaporation of nano-water droplets under realistic conditions with advanced MD water models.

2. Method

2.1. Molecular dynamics simulation

MD simulations are performed using the large-scale atomic/-molecular massively parallel simulator (LAMMPS) [29] and results are post-processed using Visual Molecular Dynamics (VMD) [30]. Evaporation on static water droplets in varying ambient temperature was simulated and compared with existing literature [31]. The TIP4P water model with a long-range coulombic solver, the particle–particle particle-mesh (PPPM) solver [32], is used to model the behavior of the water molecules in our simulation [20].

The simulation domain is illustrated in Fig. 1. Water droplet of diameter D is placed right in the middle of the domain, and sufficient O_2 and N_2 molecules are added to the domain surrounding the water droplet to model the density of air at standard atmosphere. For intra molecular interactions within O_2 and N_2 molecules, harmonic bond potential was used [33–34]. Intermolecular interactions for O_2 and N_2 are based on Lennard-Jones potential adopted from [33–34]. All simulations begin with an equilibration at isothermal-isobaric conditions (NPT) with a timestep of 0.5 fs for 100,000 timesteps, followed by a further equilibration at NPT conditions with a timestep of 1 fs for 50,000 timesteps. During equilibration, pressure was set at 1 atm, and temperature at 293 K. After equilibration at 293 K, the domain is heated to the desired temperature over 50,000 timesteps at timestep of 1 fs. Production runs at the desired temperature then commences, with simulated runtime between 6 and 40 ns. Each run was conducted as far as possible until half of the molecules in the water nanodroplet have evaporated. However, due to the long computational time required, this was not possible for all droplet sizes. The collected data shows that the trend number of water molecules evaporated over time is rather linear for all sizes and temperatures considered in this work, hence, it was concluded that there will not be significant errors for the data collected from runs that were not able to reach half-life.

In the simulations, water molecules are removed after they leave the droplet surface to ensure that the relative humidity (RH) of the surrounding air remains close to the desired RH, in this case close to 0 %. Due to the small size of the domain and the periodic boundary conditions imposed, the presence of one water

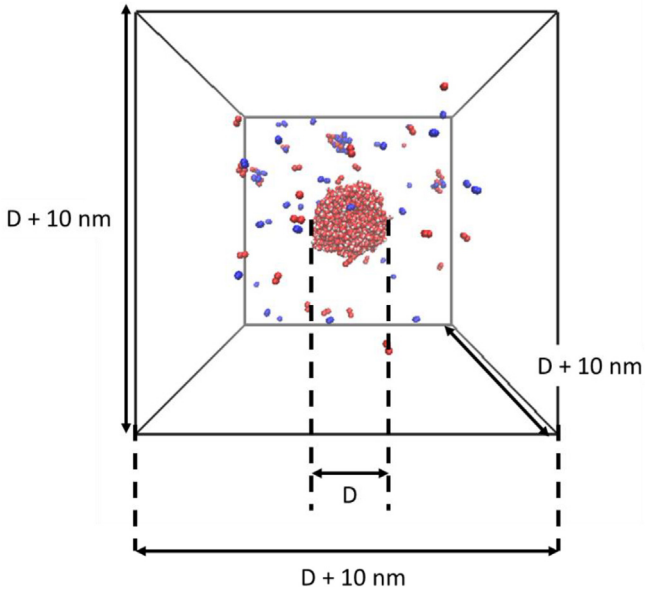


Fig. 1. Simulation domain.

molecule is enough for the air to alter the RH of the surrounding air significantly. Hence, it is essential to remove water molecules from the simulation domain periodically as it evaporates. This is done in LAMMPS via the fix evaporate command after the water molecules leave the surface of the water droplet.

The diameter of the water droplets is computed by using the density of liquid water ρ (1)g/cm^3 using the Equation (1):

$$d(cm) = \left(\frac{M_r \times n_w}{N_A\left(\frac{4}{3}\pi\right)(\rho)}\right)^{\frac{1}{3}} \times 2 \tag{1}$$

Where M_r is the molar mass of water and N_A is the avogadro's constant.

2.2. Continuum calculations

Throughout this report, MD results are compared with continuum predictions. The continuum calculations are performed using the equation for the rate of diameter change for an evaporating droplet presented by Kincaid and Langley [31,35]. For ease of reference, a summary of the continuum equations is provided herein. The rate of change of a droplet diameter dD/dt is shown in Equation (2).

$$\frac{dD}{dt} = \frac{C_1}{D} \tag{2}$$

where.

$$C_1 = -2\left(\frac{M_v}{M_m}\right)K\left(\frac{\rho_a}{\rho_d}\right)\left(\frac{\Delta p}{p_f}\right)N_u^{'} \tag{3}$$

M_v is the molecular weight of water (18 g/mol), M_m is the molecular weight of air (29 g/mol for dry air), ρ_a is the density of air (1.225 kg/m^3), ρ_d is the density of water (1000 kg/m^3).

K is the diffusivity given by the following equation:

$$K = \left(\frac{101.3}{P_a}\right)8.8 \times 10^{-10}T_k^{1.81} \tag{4}$$

Where P_a is the atmospheric pressure in kPa, T_k is the water temperature in K.

$\frac{\Delta p}{p_f}$ is given by:

$$\frac{\Delta p}{p_f} = \frac{P_s - P_v}{P_a - P_v} \tag{5}$$

P_s is the saturation pressure at the wet bulb temperature of air, P_v is the saturation pressure at the dew point or vapor pressure at the dry bulb temperature.

$Nu^{'}$ is the Nusselt number given by:

$$Nu^{'} = 2.0 + 0.60Sc^{1/3}Re^{1/2} \tag{6}$$

Where Sc is the Schmidt number given by $\frac{v}{K}$, where v is the kinematic viscosity of air (1.48e-5 m^2/s). Re is the Reynolds number given by $\frac{DV}{v}$, where V is the velocity of the droplet.

From Equation (2), the differential equation can be integrated to obtain Equation (7),

$$D^2 = D_i^2 + 2C_1t \tag{7}$$

where D_i is the initial droplet diameter and t is the time of evaporation. The diameter D of the droplet can then be calculated by stepping forward in time.

3. Results

3.1. Molecular dynamics (MD) results for 4 nm droplet

To start off, MD simulations were performed with the initial size of the water nanodroplet fixed at 1000 water molecules or 4 nm in diameter. This size was selected so that comparison can be made with the MD study performed by Desai and Kaware [31], where the evaporation behavior of 4 nm water droplet surrounded by rarified liquid vapor is studied. Few other MD works are available in literature that do atomistic simulations to study the volumetric change during the evaporation of water droplets using modern force potentials, as far as the authors are aware. Comparison between the results obtained herein are also made with continuum predictions and discussed collectively. The MD simulations were conducted for three different temperatures, 30 °C, 50 °C and 70 °C. It is assumed that the temperature of the air is in equilibrium with the temperature of the water droplet in this study. This assumption effectively reduces the effects of convective heat transfer. Hence, the dominant mode of energy transfer is the latent heat of vaporization, which is the energy required for each water molecule to escape from the water droplet surface. For a water nanodroplet, this is a fair assumption, since the small volume would mean that the temperature can equilibrate very quickly with the surroundings. However, this assumption limits the results to nanodroplets that are of temperature close to the surrounding ambient air, since if the temperature of the ambient air is significantly higher than the nanodroplets, much of the vaporization would occur during the thermal equilibration phase. RH for this first study was fixed at close to 0 %.

The temporal evolution of number of water molecules remaining in the droplet predicted by MD study from this work is shown in Fig. 2(a). As shown, the simulations were conducted as long as possible until the droplets reached half-life, which represents the state when half of the water molecules that initially exist in the droplet have been evaporated away. However, at a low temperature of 30 °C, the simulation time is prohibitively long. In addition, from the results obtained at 50 °C and 70 °C, the temporal evolution of the decrease in number of molecules within the droplet over time is rather linear. Therefore, it is concluded that there is no significant error in using the first 40 ns of evaporation for 30 °C to predict the droplet's evaporation behavior. The temporal evolution of the diameter of droplets as evaporation takes place are shown in Fig. 2(b). In the same figure, the MD results are further compared with that predicted using continuum equations.

From Fig. 2b, the evaporation of nanosized water droplets is much slower than that predicted by continuum equations. The continuum equations computed at 50 °C predicts that the water droplet should be fully evaporated in less than 5 ns. Based on our MD simulations, it could take more than 100 ns. For room temperature (30 °C), the evaporation time is expected to be even longer. Interestingly, our results showed that evaporation of water droplets in ambient air conditions seemed to be much slower than when water droplets are surrounded by rarefied liquid, as demonstrated by Desai and Kaware [31]. Desai and Kaware [31] showed that at 50 °C the evaporation rate of the 4 nm water droplet surrounded by rarified liquid is similar to that from continuum equation, which is much slower than predicted by our MD simulations.

To illustrate the evaporation process, Supplementary Video 1 (SV1) shows the video of the 4 nm water nanodroplet evaporating at 50 °C. The bottom right of the video illustrates the isometric view of the simulation domain, with the water droplet right at the middle surrounded with air molecules. The main video shows a slowed down video where two water molecules, highlighted in yellow, can be seen to escape from the surface of the water droplet

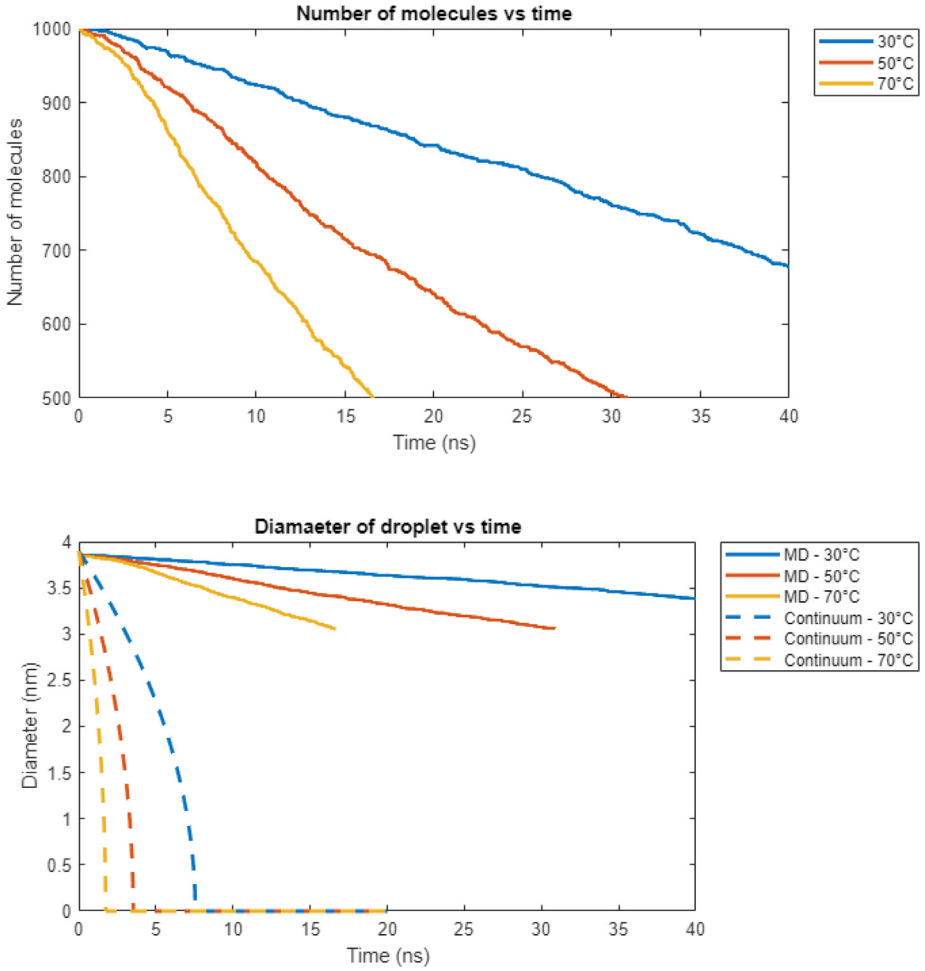


Fig. 2. MD results for water nanodroplets evaporation of a 4 nm sized droplet. (a) Temporal evolution of number of molecules remaining in droplets over time. (b) Temporal evolution of diameter of droplets over time predicted by MD (shown in solid lines) as compared to that predicted using continuum equations (shown in dashed lines).

and gets evaporated away. In an evaporation process, two key phenomenon are at play at the interface of the liquid–vapor boundary. First, the transfer of heat from surrounding gas to the liquid increases the kinetic energy of the molecules at the boundary, giving it enough momentum to escape from the attractive cohesive forces, primarily driven by intermolecular hydrogen bonds for water nanodroplets. In the simulation, it is assumed that the water droplet is always at thermal equilibrium with the surrounding in this simulation, which is a fair assumption given the small volume of nanodroplets. The second phenomenon driving the process of evaporation is the concentration difference encountered by surface liquid molecules. This is the primary phenomenon observed in this simulation, and for applications where the nanodroplets can be assumed to be at the same temperature as its surrounding. The forces at play here are the kinetic energy of the water droplets overcoming the intermolecular hydrogen bonds (H-bonds) between water molecules. It is known and can be observed from SV1, that at the surface, the number of H-bonds decreases when compared to the interior [36]. In the bulk interior, water can form up to four H bonds with neighboring water molecules [37] while at the surface, that decreases to about 3 [36]. As the kinetic energy of the water molecule exceeds the energy of the H-bond at the surface of water, the surface water molecules acquire the required energy to break free and become water vapor. This process is clearly illustrated in SV1.

Investigating the MD results in Fig. 2a carefully, the reduction of the ratio of the number of water molecules left in the droplet with time can be approximated using a linear trend. Curve fitting was therefore performed using MATLAB curve fitting toolbox [38] to fit the data collected to a linear equation of the form given in Equation (8), where A and B are curve fitting parameters. N_w is the original number of molecules within the water droplet.

$$n_w = N_w + Bt \tag{8}$$

In Equation (8), only B needs to be fitted. B physically represents the evaporation rate in #/ns. The R^2 values, calculated according to Equation (9) and the corresponding value for B for the three cases considered are given in Table 1. The fit is very good, with R^2 values very close to 1. The fitted curve is shown in Fig. 3a at the simulated time range to illustrate the fit and extended to 100 ns in Fig. 3(b) to show the predicted evaporation transient time profile for the 4 nm droplet.

$$R^2 = \frac{\sum_{i=1}^{n}\left(\hat{y}_i - \bar{y}\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{9}$$

In summary, the current MD simulations confirm that for a nanometer water droplet, continuum calculations can no longer be used to approximate the evaporation rate. As can be seen in Fig. 2, the continuum prediction over-predicts the evaporation rate of nanosized water droplets significantly. While simple curve fitting can predict the evaporation trajectory of a 4 nm sized droplet, it will be more meaningful if the evaporation behavior of nan-

Table 1
Results of curve fitting to determine the evaporation rate ***B*** for 4 nm sized water droplets.

Initial number of molecules in droplet (N_w)	Initial surface area of droplet A nm^2	Initial diameter of droplet D (nm)	Temperature T°C	B (#/ns)	R^2
1000	12.57	4	30	−7.90	0.998
1000	12.57	4	50	−17.35	0.989
1000	12.57	4	70	−30.64	0.991

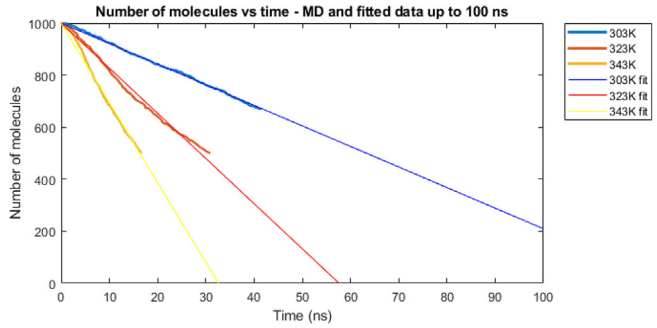


Fig. 3. The predicted temporal evolution of number of water molecules in droplet over time from the linear fit.

odroplets of varying size can be predicted. Since the computational cost of each MD simulation is very expensive, it is not practical to determine the evaporation rate of a nano-droplet for any condition using MD. Hence, this work aims to provide a framework methodology that can be used to determine the evaporation rate of water nanodroplets of other sizes, at other temperatures, using a data driven approach. Here, 2 other sets of MD simulations are performed at different sizes (5nmand6nm) and temperature conditions (30 °C, 50 °C and 70 °C) to build a data base that can be used to estimate the evaporation rate for water nanodroplets. This model is then validated on a 8 nm sized water droplet. This framework forms the foundation that such data driven method is possible and hope to inspire other researchers to contribute to the database so that the evaporation rate at other conditions, for example humidity, proximity of other droplets, ambient temperature gradient, can also be predicted.

3.2. MD results for droplets of other sizes

To scale up the prediction of the temporal evolution of the droplet size during evaporation to different sizes, the same fitting process was also carried out for the 5 and 6 nm droplets. The data obtained is presented in Fig. 4 and Table 2.

As with the 4 nm droplets, linear fitting works very well for 5 nm droplets. However, some deviations are seen in the 6 nm sized droplet. As the droplets get larger, it can be observed that

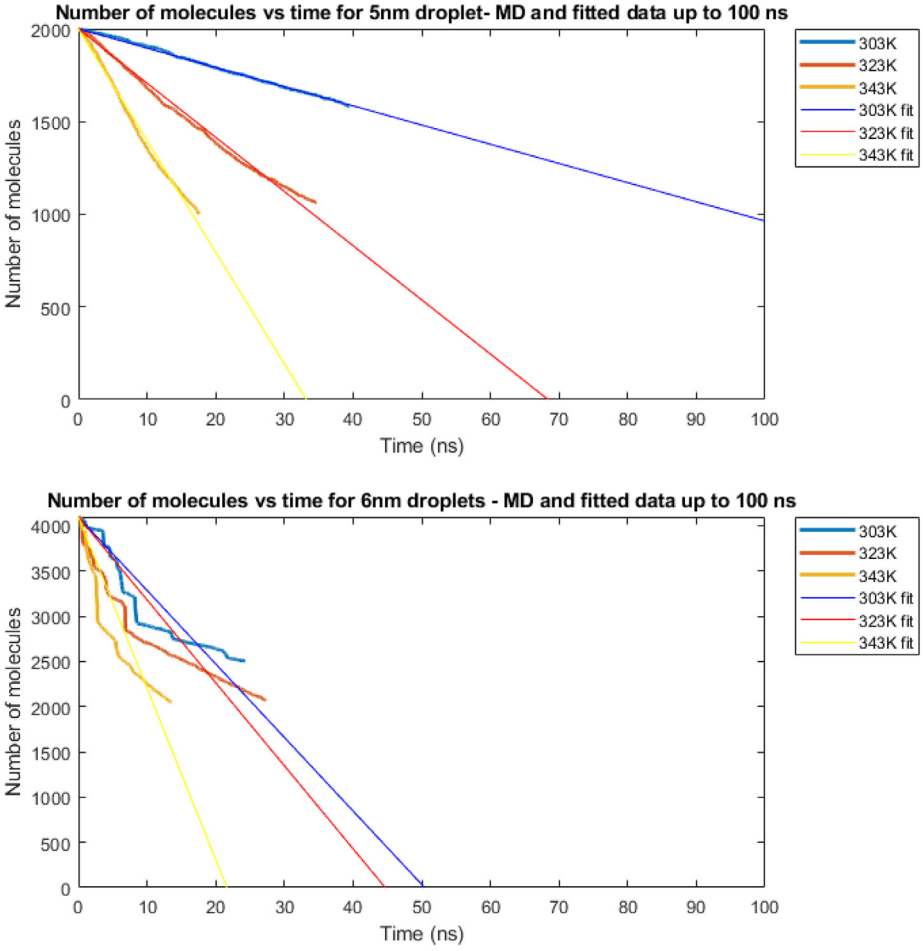


Fig. 4. The predicted temporal evolution of number of water molecules in droplet over time using MD and linear fit for 5 nm (top) and 6 nm (bottom) droplet.

Table 2
Parameters of the fitted model for 5 and 6 nm droplets.

Initial number of molecules in droplet (N_w)	Initial surface area of droplet A nm^2	Initial diameter of droplet D (nm)	Temperature T °C	B (#/ns)	R^2
2000	19.63	5	30	−10.36	0.997
2000	19.63	5	50	−29.24	0.990
2000	19.63	5	70	−60.24	0.991
4096	28.27	6	30	−81.23	0.814
4096	28.27	6	50	−91.66	0.574
4096	28.27	6	70	−189.33	0.697

the number of molecules remaining in the droplets over time are more haphazard and deviates from a linear trend. Upon further investigation, the authors attributed this to the larger intermolecular forces of attraction at play when the droplet size is larger. Larger droplets have a larger center of mass which holds the water molecules together tighter, as governed by the newton's law of universal gravitation. However, the surface area for a larger droplet is also significantly larger than a smaller droplet. Hence, while more water molecules are at the surface, thus having a smaller number of H bonds which increases its chance of being evaporated, the larger intermolecular attractive force is holding the water molecules tighter to the droplet. These competing forces are what result in the more haphazard profiles observed for bigger droplets. Hence, it can be concluded that the linear model adopted in this contribution is expected to be more inaccurate for larger droplets, and a more complicated curve fitting procedure, or more sophisticated machine learning models, are expected to be required as droplet size gets larger. The difference in behavior between a smaller 4 nm water droplet as compared to a larger 8 nm water droplet is shown in SV2. The 8 nm water droplet is larger and the water molecules are held tighter in a spherical shape, whilst a 4 nm water droplet is looser.

For demonstration purposes, this contribution decides to stick with the linear model for its simplicity, ease of understanding and cheap computational cost. It will be shown next that while the linear fit for a 6 nm droplet may not be very accurate, the resulting model built based on the data generated from water nanodroplets of size 4, 5 and 6 nm can predict the evaporation trajectory of an 8 nm droplets to its half-life with good accuracy.

3.3. Data driven mathematical model from MD results

To scale up the prediction of evaporation of water nanodroplets to different sizes, this contribution makes use of a distinct pattern observed from Table 2. Evaporation rate B increases in magnitude with temperature, which is expected. However, the rate at which B increases in magnitude with T (named m in this work) is observed to increase with initial area A as observed in Fig. 5. Plotting m with the initial surface area of the droplet, a quadratic trend is observed as seen in Fig. 6. The fitted quadratic curve using data obtained from MD simulations involving 4, 5 and 6 nm droplets, collected every 0.02 ns (total of about 18,000 data points) is given in Equation (10). Therefore, it is suggested that the fitted relationship shown in Equation (10) can be used to calculate m for different initial area of the water nanodroplets. This can then be used to calculate the evaporation rate B at different temperature and initial droplet sizes using Equation (11). Hence, any user can predict the evaporation rate B in #/ns of a nanodroplet with initial diameter d using Equation (11). Firstly, one would compute the approximated initial surface area A, using $A = \frac{\pi d^2}{4}$. Then with the ambient temperature T, the evaporation date can then be easily compute.

$$m = -0.0043A^2 + 0.037A \tag{10}$$

$$B = \left(-0.0043A^2 + 0.037A\right)T \tag{11}$$

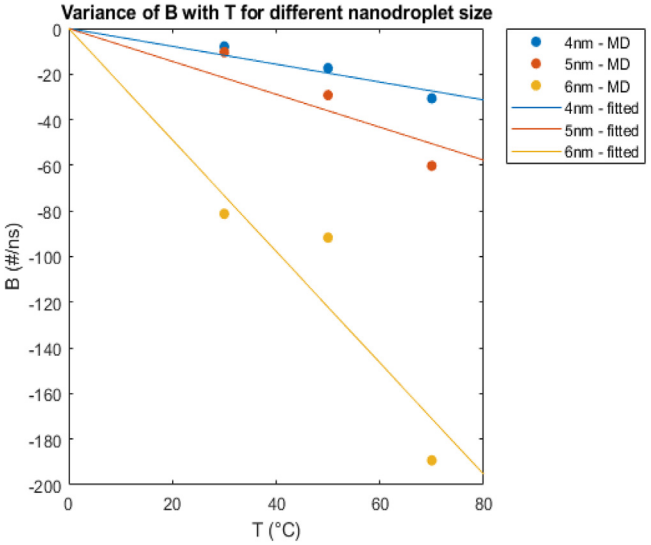


Fig. 5. Plot showing how B varies with T for different nanodroplet size. Note that the gradient of the B vs T graph is denoted as m in this contribution.

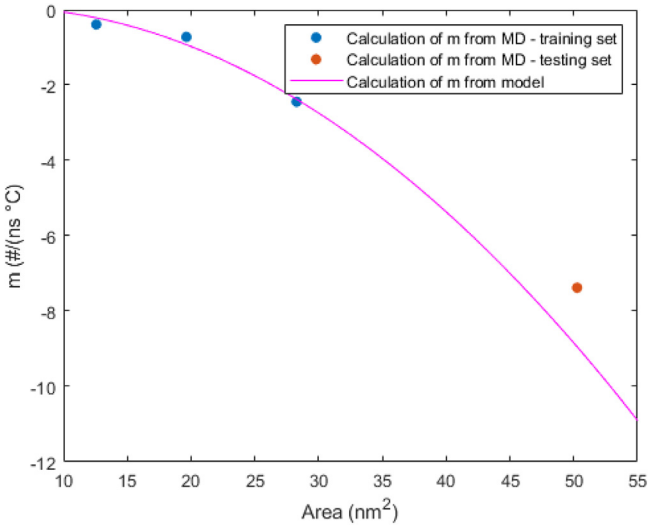


Fig. 6. Plot showing how m, the rate of change of B with T, changes with initial surface area of droplet.

To test the model, we collected an additional set of MD simulations with an 8 nm sized droplet. The model given by Equation (11) is used to approximate the evaporation rate B of this 8 nm sized droplet. The predicted m using Equation (11) for 8 nm sized droplet with area A = 50.27 nm^2 is −8.95 $\frac{\#}{ns^\circ C}$. MD simulations for the evaporation of 8 nm sized droplet is then carried out at temperatures T = 30 °C, 50 °C and 70 °C. The resulting m measured

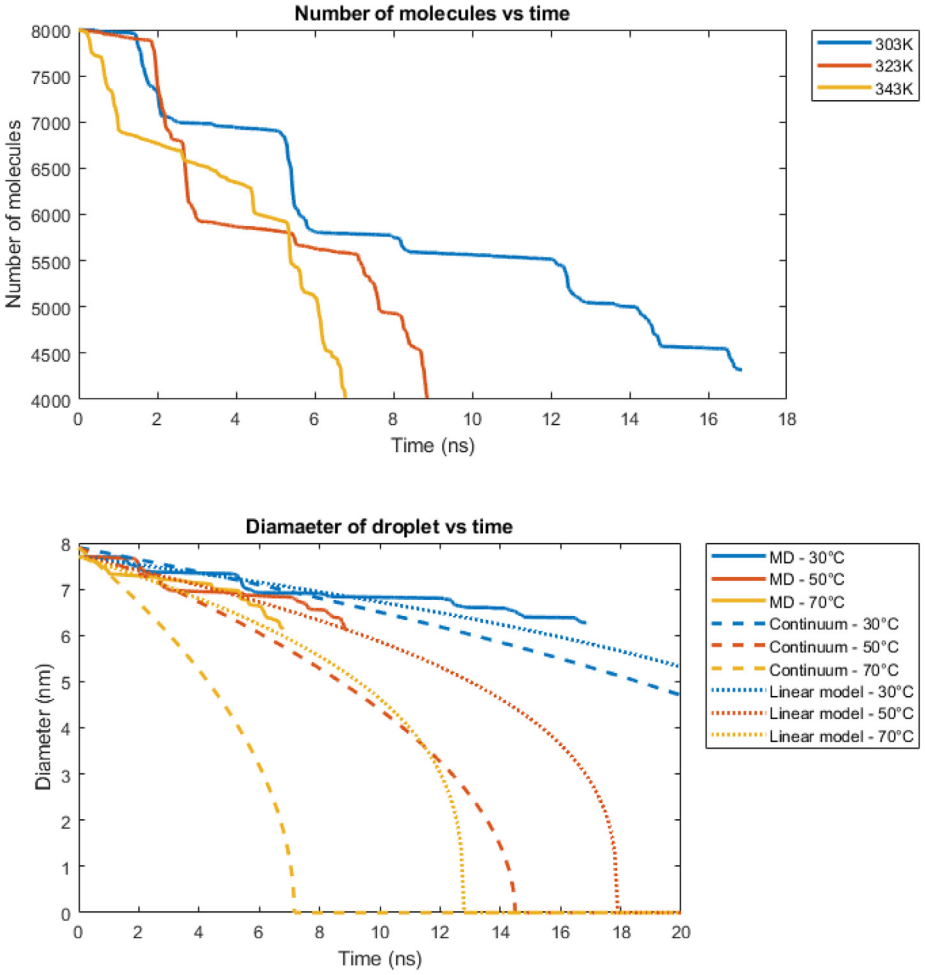


Fig. 7. Temporal evolution of 8 nm droplet predicted using full atomistic MD simulation, continuum predictions and linear model. The linear model can predict the temporal evolution of the 8 nm droplet in very close agreement with MD simulation, at a fraction of its cost.

from MD simulations are plotted as an orange circle in Fig. 6 as $= -7.38\ \frac{\#}{ns^{\circ}C}$, and the resulting evaporation profiles in terms of number of water molecules and diameter of the droplet is shown in Fig. 7. The prediction from continuum calculations, and the linear data-driven model according to Equation (11) are shown together in Fig. 7b. While the continuum model consistently overpredicts the evaporation rate, the linear model can predict the temporal trajectory of the reduction in diameter over time for 8 nm sized droplet very well. This shows that the data-driven linear model does have the capability to accurately predict the evaporation behavior of a nano-sized water droplets without having to run further MD simulations.

4. Conclusion

There are two main contributions from this paper. First, the MD study confirms that the d2-law based continuum predictions cannot be used to predict the rate of evaporation for suspended nanodroplets. The continuum predictions significantly overpredicts the rate of evaporation. Hence, applications which require accurate knowledge of nanodroplets evaporation rate, such as nano ink-jet printing or nanofabrication, or even macro-scale applications such as better weather forecasting, can benefit from this knowledge. Second, data-driven methods can be the key to effectively predict the evaporation rate of nano water droplets at different conditions. While MD is recognized as an accurate way to predict nano-scale phenomenon, the high cost associated with it is limiting its application to real-world problems. In this paper, an alternative data-driven model that is obtained through curve fitting is proposed instead to predict the evaporation trajectory of nano droplets to a good level of accuracy. While it is recognized that the current model is highly simplified, with larger datasets available, more complicated, and powerful data-driven models can be built to predict the evaporation trajectory of nanodroplets in different conditions such as initial size, temperature, humidity, or proximity to other droplets. The authors will make the MD results obtained through this study publicly available and hope to inspire further research in this field (url: https://github.com/elisaaym/Water_nanodroplets_evaporation).

CRediT authorship contribution statement

Elisa Y.M. Ang: Methodology, Conceptualization, Software, Validation, Formal analysis, Writing – original draft. **Peng Cheng Wang:** Conceptualization, Writing – review & editing, Supervision. **William Toh:** Writing – review & editing, Supervision. **Teng Yong Ng:** Writing – review & editing, Supervision.

Data availability

Data is hosted in github with link provided in manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to express their appreciation to Nanyang Technological University (NTU) High Performance Computing Centre (HPCC) and National Supercomputing Centre (NSCC) Singapore for providing all the necessary supercomputing resources and support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2022.121034.

References

[1] M.H. Biroun, M. Rahmati, R. Tao, H. Torun, M. Jangi, Y. Fu, Dynamic Behavior of Droplet Impact on Inclined Surfaces with Acoustic Waves, Langmuir 36 (2020) 10175–10186.
[2] A. Li, H. Li, Z. Li, Z. Zhao, K. Li, M. Li, Y. Song, Programmable droplet manipulation by a magnetic-actuated robot, Sci. Adv. 6 (2020) 1–7.
[3] A.S. Rana, D.A. Lockerby, J.E. Sprittles, Lifetime of a Nanodroplet: Kinetic Effects and Regime Transitions, Phys. Rev. Lett. 123 (2019).
[4] A. Tiribocchi, A. Montessori, M. Lauricella, F. Bonaccorso, S. Succi, S. Aime, M. Milani, D.A. Weitz, The vortex-driven dynamics of droplets within droplets, Nat. Commun. 12 (2021) 1–10.
[5] S. Gant, J. Weil, L. Delle Monache, B. McKenna, M.M. Garcia, G. Tickle, H. Tucker, J. Stewart, A. Kelsey, A. McGillivray, R. Batt, H. Witlox, M. Wardman, Dense gas dispersion model development and testing for the Jack Rabbit II phase 1 chlorine release experiments, Atmos. Environ. 192 (2018) 218–240.
[6] H. W. M. Witlox, M. Harper, R. Pitblado, Validation of phast dispersion model as required for USA LNG siting applications. *12th Topical Conference on Gas Utilization 2012 - Topical Conference at the 2012 AIChE Spring Meeting and 8th Global Congress on Process Safety*. **31**, 263–275 (2012).
[7] L. Morawska, J. Cao, Airborne transmission of SARS-CoV-2: The world should face the reality, Environ. Int. 139 (2020).
[8] S. Kumar, Insight on the evaporation dynamics in reducing the COVID-19 infection triggered by respiratory droplets, Cite as: Phys. Fluids. 33 (2021) 72004.
[9] Y. Drossinos, T.P. Weber, N.I. Stilianakis, Droplets and aerosols: An artificial dichotomy in respiratory virus transmission, Health Sci. Rep. 4 (2021) 1–7.
[10] J.S. Walker, J. Archer, F.K.A. Gregson, S.E.S. Michel, B.R. Bzdek, J.P. Reid, Accurate Representations of the Microphysical Processes Occurring during the Transport of Exhaled Aerosols and Droplets, ACS Cent. Sci. 7 (2021) 200–209.
[11] H. Li, F.Y. Leong, G. Xu, Z. Ge, C.W. Kang, K.H. Lim, Dispersion of evaporating cough droplets in tropical outdoor environment, Phys. Fluids 32 (2020) 1–11.
[12] J. Reutzsch, C. Kieffer-Roth, B. Weigand, A consistent method for direct numerical simulation of droplet evaporation, J. Comput. Phys. 413 (2020).
[13] Y. Niimura, K. Hasegawa, X.-D. Wang, Evaporation of droplet in mid-air: Pure and binary droplets in single-axis acoustic levitator, PLoS One 14 (2) (2019) e0212074.
[14] S. Chaudhuri, S. Basu, P. Kabi, V.R. Unni, A. Saha, Modeling the role of respiratory droplets in Covid-19 type pandemics, Phys. Fluids 32 (2020).
[15] R. Hołyst, M. Litniewski, D. Jakubczyk, Evaporation of liquid droplets of nano- and micro-meter size as a function of molecular mass and intermolecular interactions: Experiments and molecular dynamics simulations, Soft Matter 13 (2017) 5858–5864.
[16] E.Y.M. Ang, W. Toh, J. Yeo, R. Lin, Z. Liu, K.R. Geethalakshmi, T.Y. Ng, A review on low dimensional carbon desalination and gas separation membrane designs, J. Memb Sci. 598 (2020).
[17] A. Hospital, J.R. Goñi, M. Orozco, J.L. Gelpí, Molecular dynamics simulations: Advances and applications, Adv. Appl. Bioinforma. Chem. 8 (2015) 37–47.
[18] L.N. Long, M.M. Micci, B.C. Wong, Molecular dynamics simulations of droplet evaporation, Comput. Phys. Commun. 96 (1996) 167–172.
[19] E.S. Landry, S. Mikkilineni, M. Paharia, A.J.H. McGaughey, Droplet evaporation: A molecular dynamics investigation, J. Appl. Phys. 102 (2007).
[20] H.W. Horn, W.C. Swope, J.W. Pitera, J.D. Madura, T.J. Dick, G.L. Hura, T. Head-Gordon, Development of an improved four-site water model for biomolecular simulations: TIP4P-Ew, J. Chem. Phys. 120 (2004) 9665.
[21] A.C.T. van Duin, S. Dasgupta, F. Lorant, W.A. Goddard, ReaxFF: A reactive force field for hydrocarbons, J. Phys. Chem. A 105 (2001) 9396–9409.
[22] D. Ju, L. Huang, K. Zhang, M. Ye, Z. Huang, G. Yi, Comparison of evaporation rate constants of a single fuel droplet entering subcritical and supercritical environments, J. Mol. Liq. 347 (2022).
[23] L. Consolini, S.K. Aggarwal, S. Murad, A molecular dynamics simulation of droplet evaporation, Int. J. Heat Mass Transf. 46 (2003) 3179–3188.
[24] B.-B. Wang, X.-D. Wang, M. Chen, J.-L. Xu, Molecular Dynamics Simulations on Evaporation of Droplets with Dissolved Salts. *Entropy 2013, Vol. 15, Pages 1232-1246*. **15**, 1232–1246 (2013).
[25] L. Ruiz Pestana, T. Head-Gordon, Evaporation of Water Nanodroplets on Heated Surfaces: Does Nano Matter?, ACS Nano 16 (2022) 3563–3572
[26] S. Coldrick, A. Kelsey, M.J. Ivings, T.G. Foat, S.T. Parker, C.J. Noakes, A. Bennett, H. Rickard, G. Moore, Modeling and experimental study of dispersion and deposition of respiratory emissions with implications for disease transmission, Indoor Air 32 (2022), https://doi.org/10.1111/INA.13000.
[27] J.C.S. Kadupitiya, F. Sun, G. Fox, V. Jadhao, Machine learning surrogates for molecular dynamics simulations of soft materials, J. Comput. Sci. 42 (2020).
[28] D. Kochkov, J.A. Smith, A. Alieva, Q. Wang, M.P. Brenner, S. Hoyer, Machine learning–accelerated computational fluid dynamics, PNAS 118 (2021), https://doi.org/10.1073/PNAS.2101784118/SUPPL_FILE/PNAS.2101784118.SAPP.PDF.
[29] S. Plimpton, Fast Parallel Algorithms for Short-Range Molecular Dynamics, J. Comput. Phys. 117 (1) (1995) 1–19.
[30] W. Humphrey, A. Dalke, K. Schulten, VMD: Visual molecular dynamics, J. Mol. Graph. 14 (1996) 33–38.
[31] S. Desai, R. Kaware, Computational modeling of nanodroplet evaporation for scalable micro-/nano-manufacturing, IIE Transactions (Institute of Industrial Engineers). 44 (2012) 568–579.
[32] R.W. Hockney, J.W. Eastwood, Computer Simulation Using Particles, (CRC Press, 2021.
[33] S.K. Thapa, N.P. Adhikari, A Molecular Dynamics Study of Oxygen Gas In Water At Different Temperatures, Int. J. Mod Phys B 27 (08) (2013) 1350023.
[34] G.M. Meconi, R. Zangi, Adsorption-induced clustering of CO2on graphene, PCCP 22 (2020) 21031–21041.
[35] D.C. Kincaid, T.S. Longley, Water droplet evaporation and temperature model, Trans. Am. Soc. Agric. Eng. 32 (1989) 457–463.
[36] F.S. Cipcigan, V.P. Sokhan, A.P. Jones, J. Crain, G.J. Martyna, Hydrogen bonding and molecular orientation at the liquid–vapour interface of water, PCCP 17 (2015) 8660–8669.
[37] A.K. Giri, F. Teixeira, M.N.D.S. Cordeiro, Structure and kinetics of water in highly confined conditions: A molecular dynamics simulation study, J. Mol. Liq. 268 (2018) 625–636.
[38] Inc. The MathWorks, Curve Fitting Toolbox (2020).